

Squash Ball Detector

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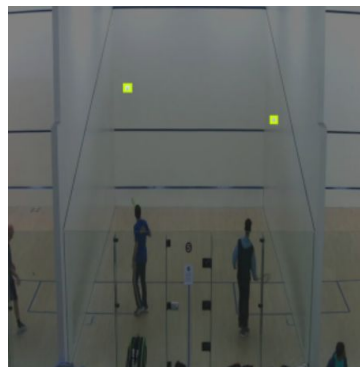
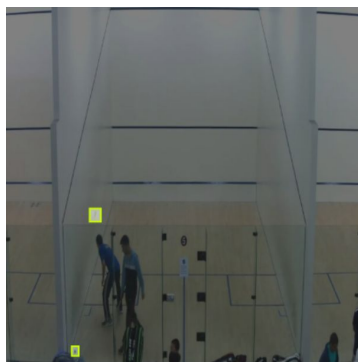
Overview

- Task: Given an image of a squash court, train a CV model to be able to accurately predict the location of the ball.
- Challenge: Unlike other sports, in squash the ball is much smaller (only 40mm diameter) and moves much faster. This, combined with images also containing players moving at high speeds nearby, makes the ball tough to identify

Dataset (Downloaded from RoboFlow)

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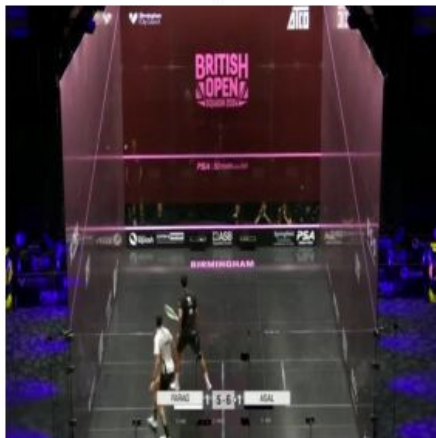
- 424 labelled images
- 1+ balls labelled per image
- Court:
wooden ground,
white walls



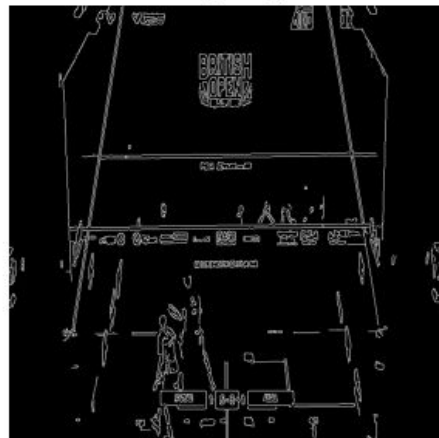
Canny Edge Filter

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- Optimize model with Canny Edge Filter
- Remove “obviously” incorrect edges (i.e. straight lines)

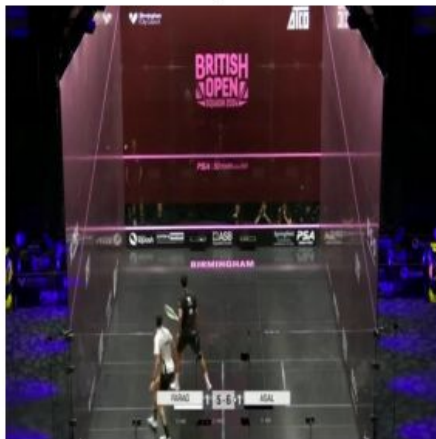
Original



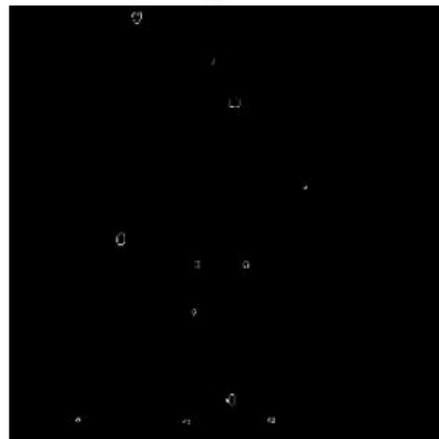
Canny Edges



Original

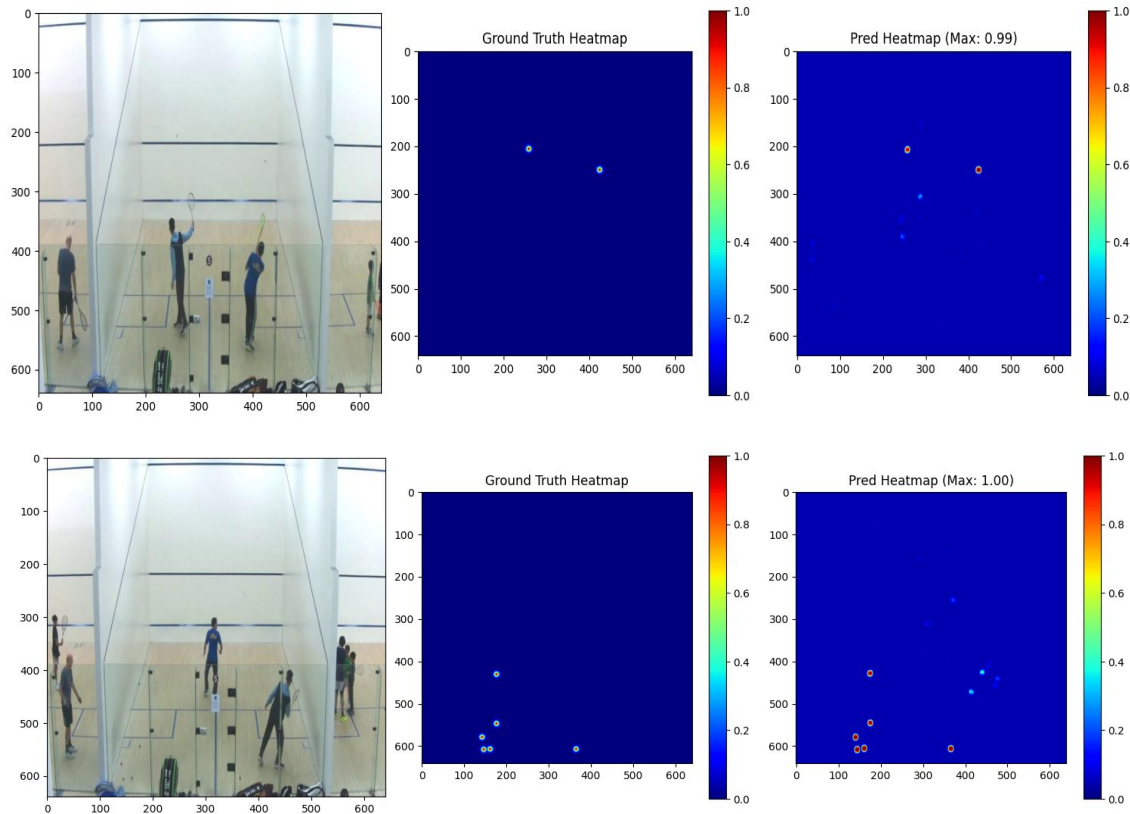


Round Edges Features



Model

- Used an Encoder-Decoder Model (w/ downsampling, then upsampling)
- Generated a heatmap
- Loss Function: Modified Focal Loss
- Better to predict false positives than false negatives
- Parameters: Alpha, Beta, Ball Position weight

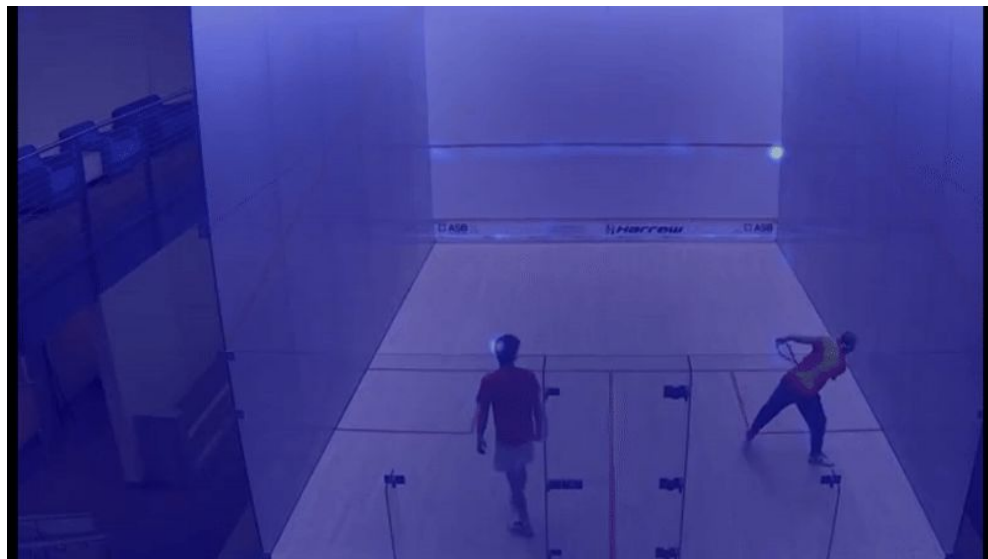


Kalman Filter

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- A Kalman Filter smooths noisy detections from model
- Maintains an accurate ball position even when individual frames have poor detections, motion blur, or partial occlusions.

Live Demo



Future Steps

1. Train and test on more diverse datasets (different style courts, color balls, camera angles, etc.)
2. Use real-time inference to train a Squash “Referee” -> learn how to track the points tally and combine ball tracker with player identifier to learn to classify obstruction calls